We're about to build the world *again.*

Cleaner energy can change how a building runs. It cannot undo the carbon already released to make it.



+55%

projected growth in global building floor area, 2022–2050.

Material choices made now will shape that world for decades.

ETC, Zero-Carbon Buildings (2025), pp. 7, 10 · UNEP (2026)

Generated illustration



Generated illustration

Build with material that grows and stores carbon.
Make places that become more alive with time.

The obstacle is production.

Natural forms vary. Living growth keeps changing.
A repeatable process has to adapt.



Different parts. One process to develop.

Bower One / digital component studies

Variability is the essence of nature and the enemy of industry;
Bower's bet is to make variability manufacturable.

Biological materials: UNEP, Building Materials and the Climate (2023). Carbon benefit depends on sourcing and the building's whole life; Bower's performance is not yet measured.


bower
Buildings that grow.
And the means to make many.
bowerbuild.org/designx/5
Architectural concept / November 2026
03

More alive with *time.*

To add: Clay's plant-growth simulation.



01 Structure

A room from the beginning.



02 Establishment

The plants find their place.



03 Maturity

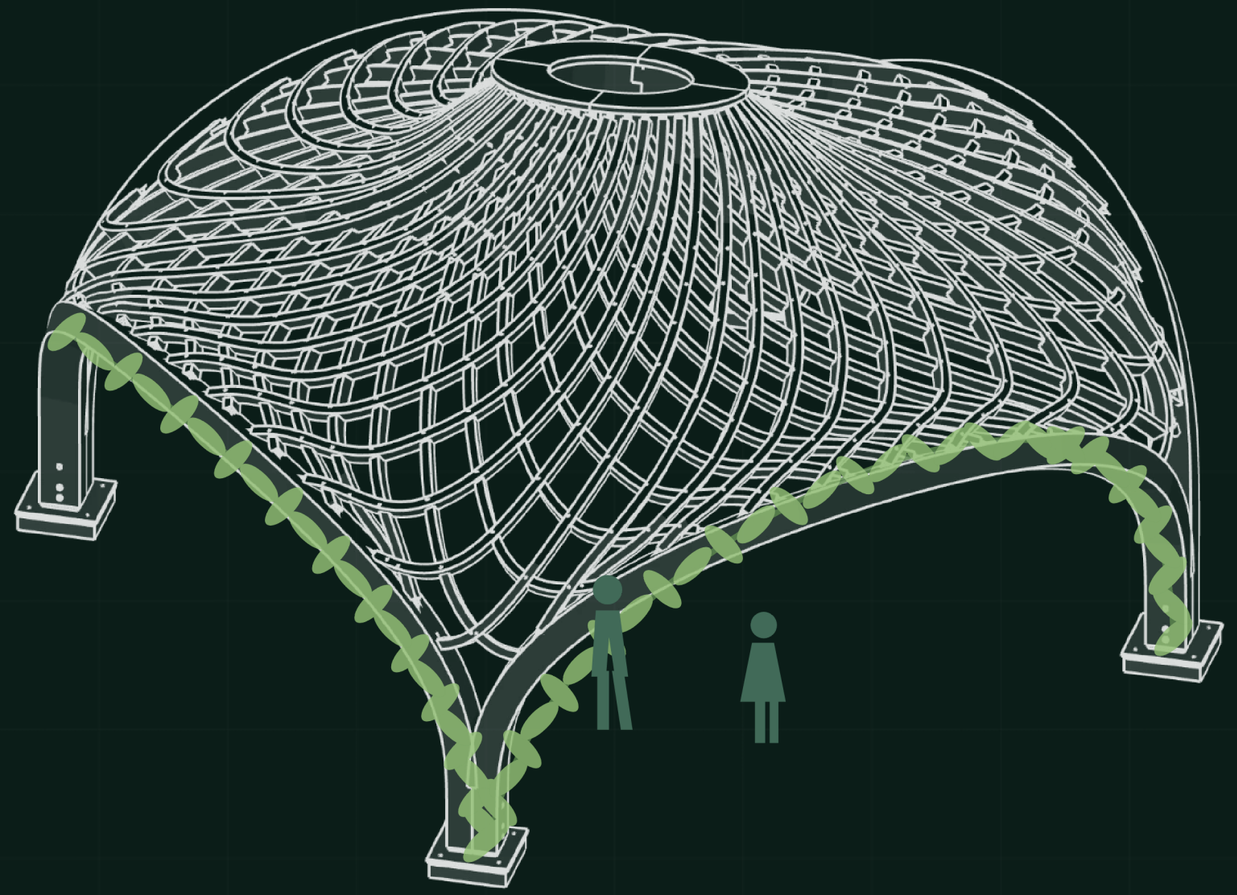
The garden transforms the room.

Illustrative growth states

A timber structure. Designed for life.

Bower assembly

To add: Timber members assemble into the Bower. Plants then colonise the frame.



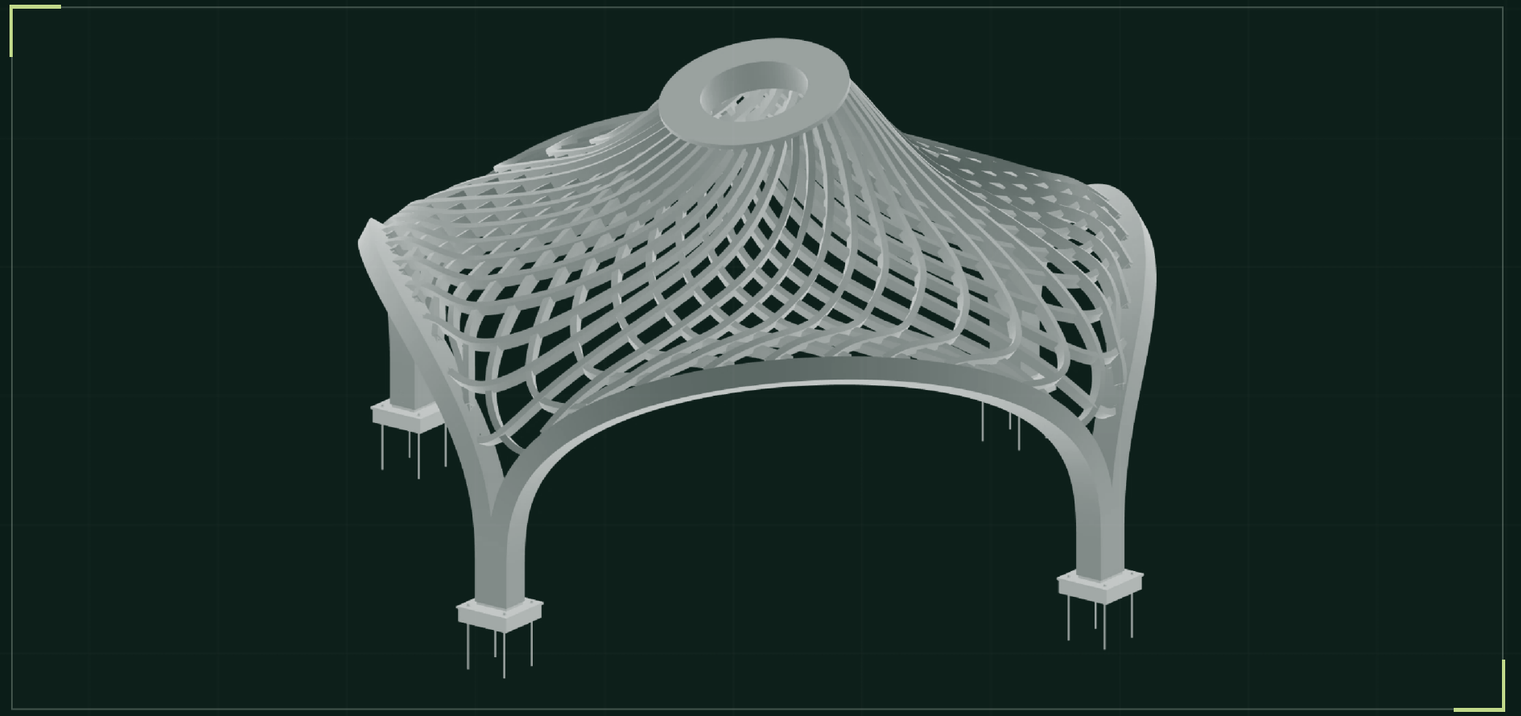
Digital assembly study. The engineered timber frame carries the structure; planting colonises it.

From form to fabrication.

Turn a changing form into parts, checks and fabrication instructions.

The engine diagram

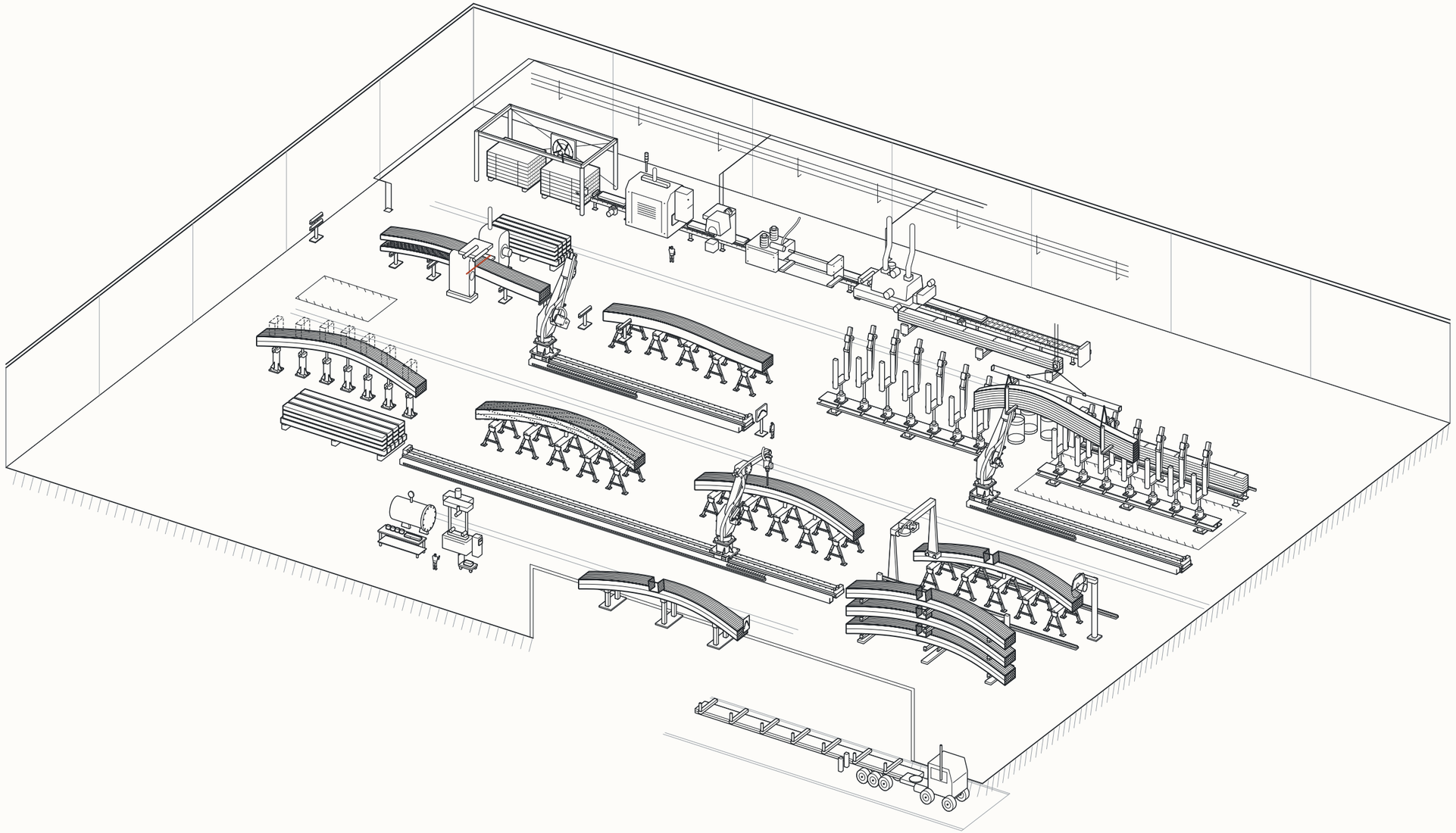
To add: Daniel’s geometry, checks and fabrication workflow. Show how they connect.



Bower One R13 · 939 model parts · 92 primary timbers

Design-to-manufacture engine in development. Existing R13 model shown; current engine diagram to follow.

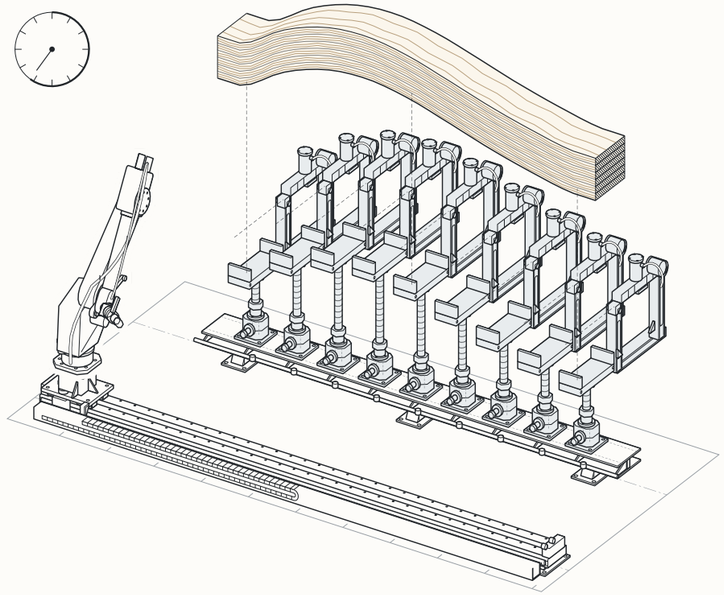
Different parts. A repeatable process.



Factory development study. Process arrangement, not a validated floor plan. Proposed line; performance and process assumptions require testing.

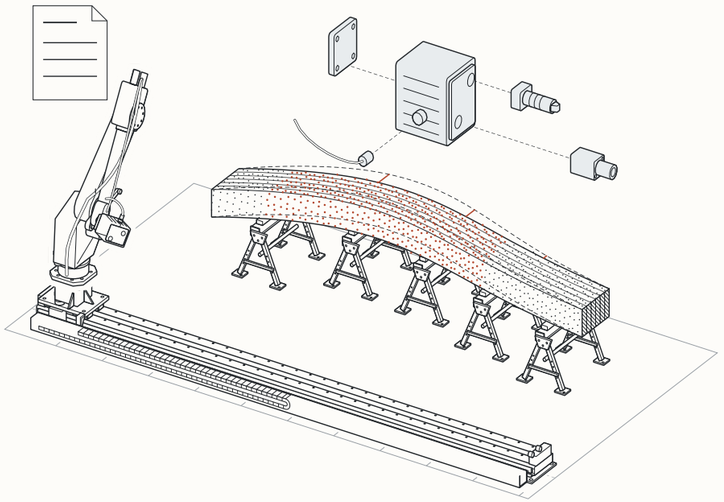
Form. Measure. Machine.

Form



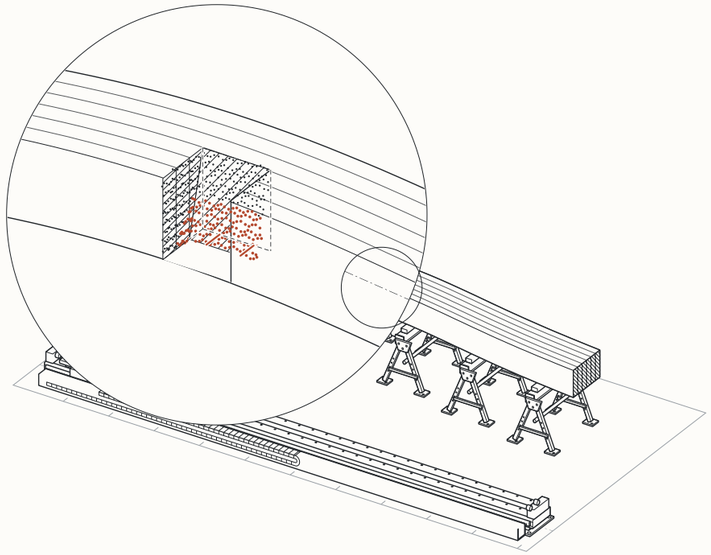
Robot-set pins guide the form.

Measure



A scan compares the rib with its target.

Machine



Joint geometry becomes a cutting sequence.

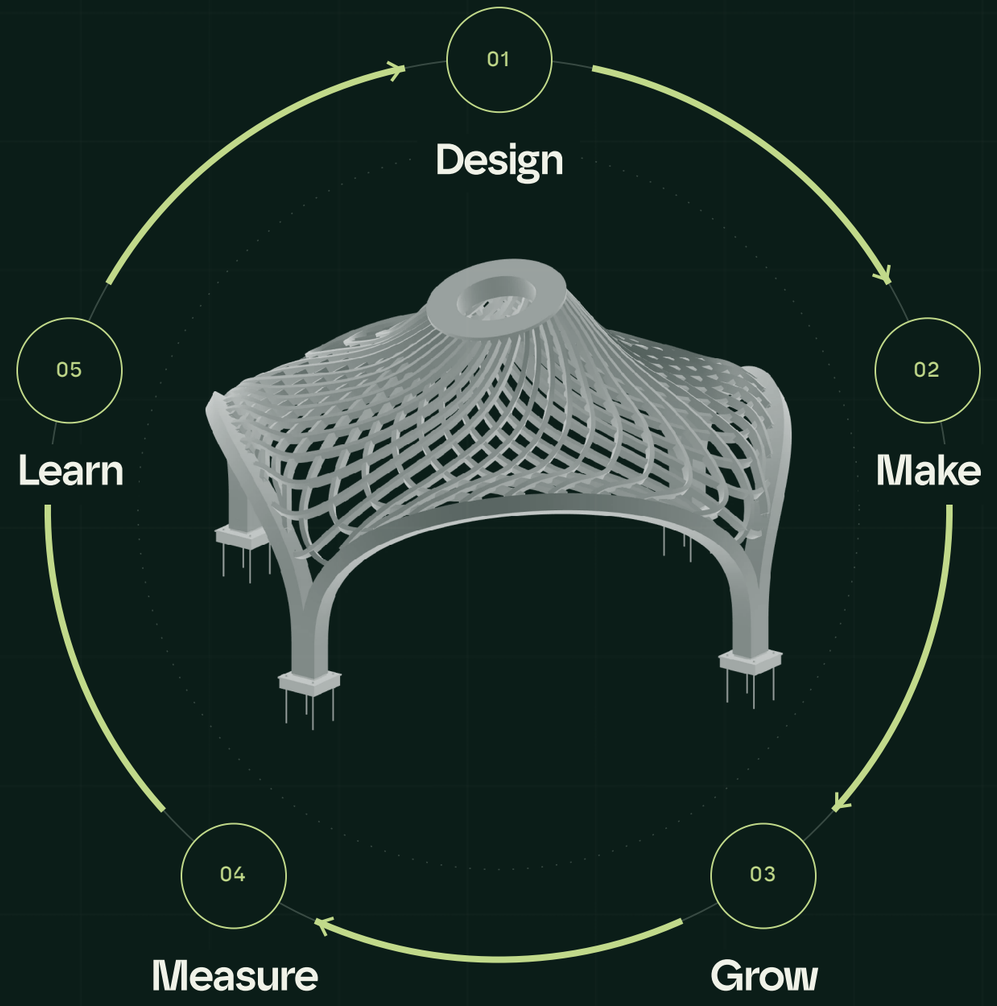
Factory development studies. Proposed process, not an operating line. Scan deviations are illustrative. Methods and assumptions in Drawing notes.

Turn each build into better design rules.

↻ THE NEXT CYCLE

A better starting point. Every time.

What the last Bower teaches the next.

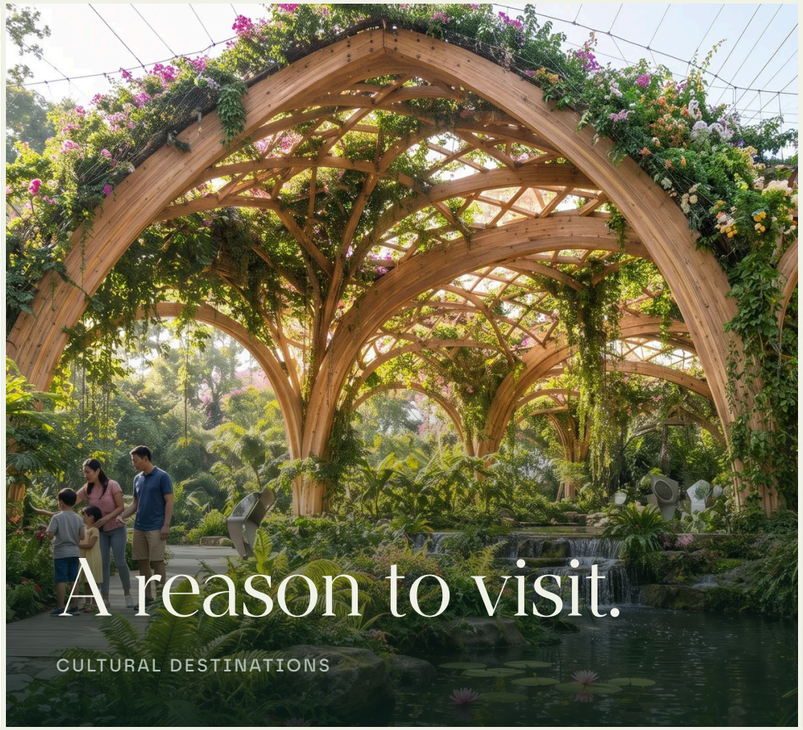


Design → Make → Grow → Measure → Learn → better design.

Intended learning system: fabrication and growth data would update design rules. Factory and garden measurement are not yet deployed.

The first *customers.*







In discussion with

eden project

The Newt in Somerset

25h

Active conversations, not signed commissions. Images are Bower concepts, not work for these organisations.

Sell a Bower.
Develop the system.

Each early commission should create a place people want—and knowledge the next build can reuse.

Siting study

£65k

indicative

Design development

£100–250k

proposal-specific

Commissioned Bower



Build terms by contract.

Reusable design rules

Tested fabrication workflows

Production tooling

Intended commercial model / indicative proposal fees. Capability development and reinvestment depend on project economics and additional finance.

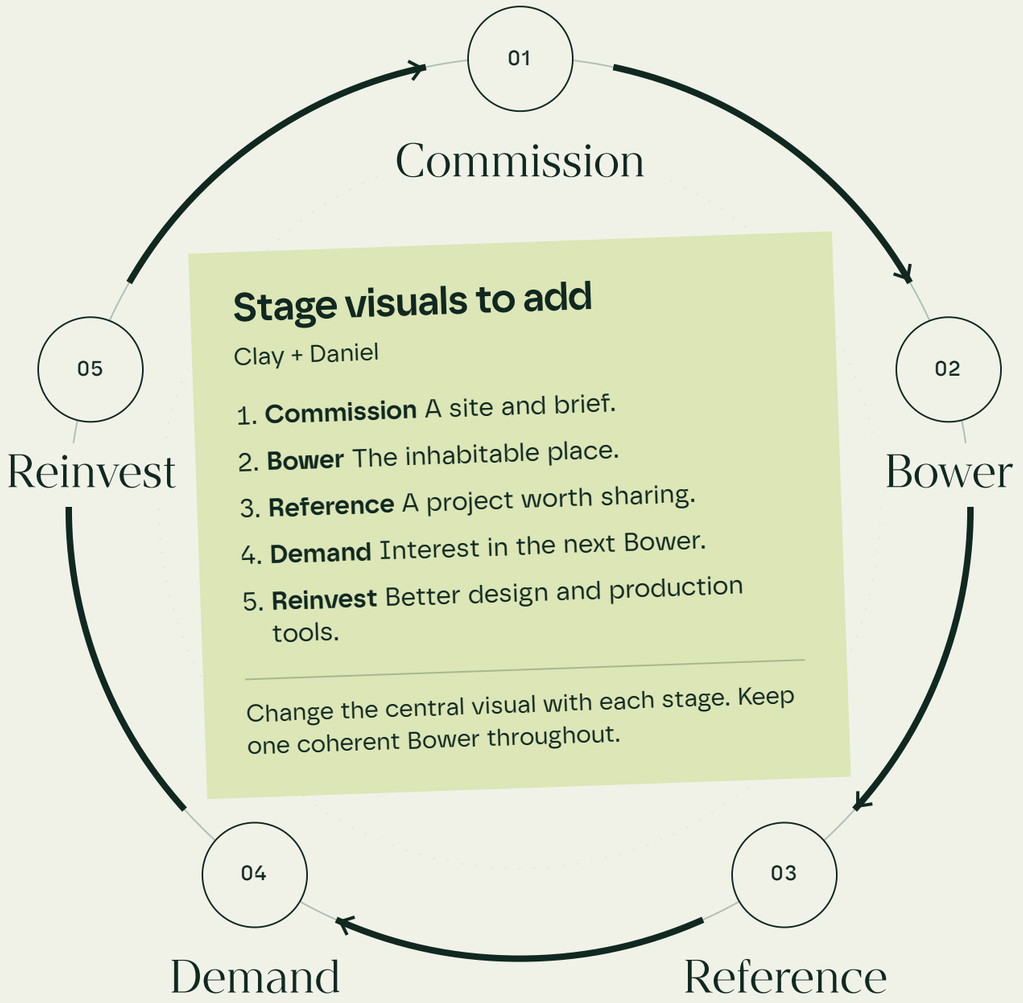
Built proof creates *demand.*

↻ THE NEXT CYCLE

The next place starts with more behind it.

Experience, references and stronger capability.

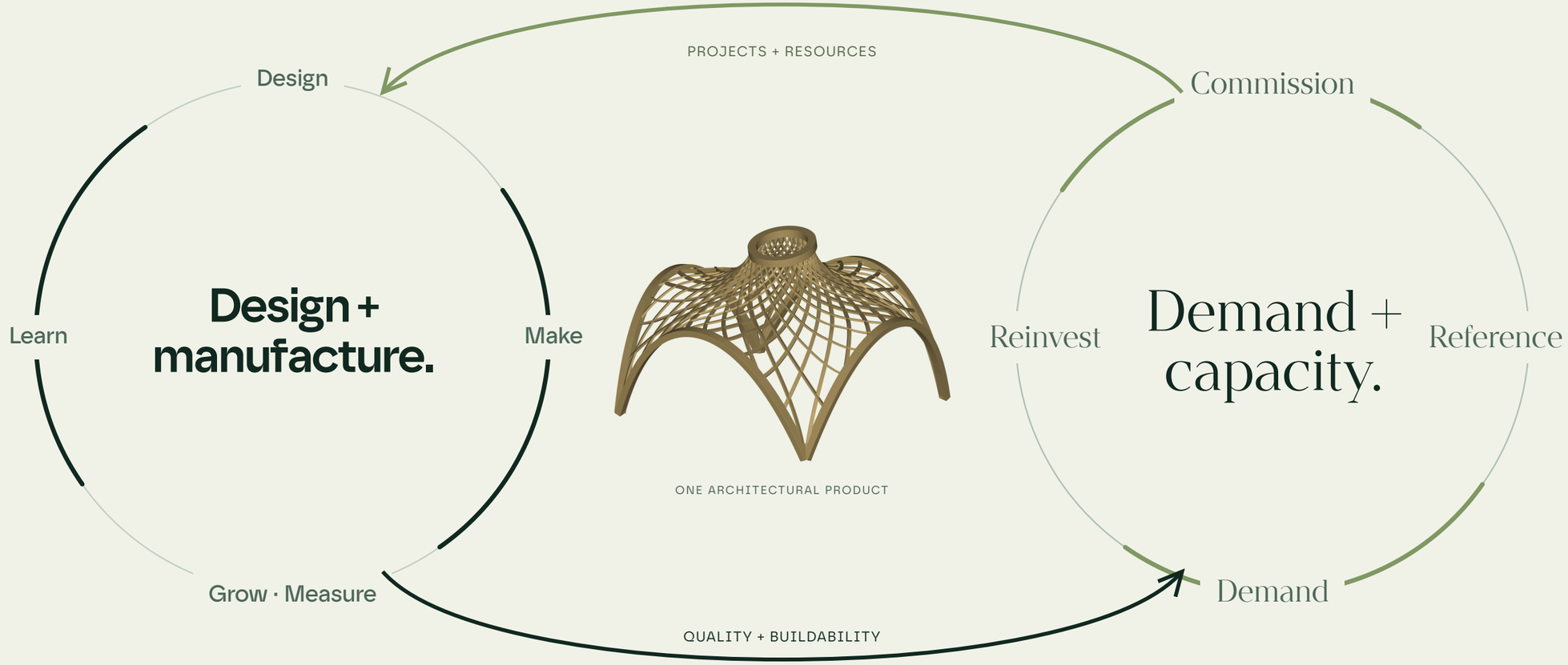
Commission → Bower → Reference → Demand → Reinvestment.



Intended commercial model. Reinvestment depends on project economics and finance.

From commissions to production capability.

First commissions test desirability and buildability.
A repeatable production system is how the ambition reaches beyond them.



Projects support learning. Learning improves projects.

The long-term objective is repeatable production. First commissions test the product, process and demand; the factory remains in development.

Architecture. Ecology.
Software. Manufacturing.



Clay Seifert

Architecture.
Computation.
Living systems.

UT Austin / B.Arch



Daniel Guerra

Software.
Fabrication.
Project delivery.

Forsite Studio / project experience

A room is the beginning.

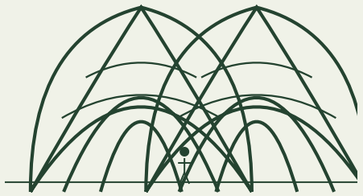
Prove the room. Develop the system. Explore what it could make possible.

01



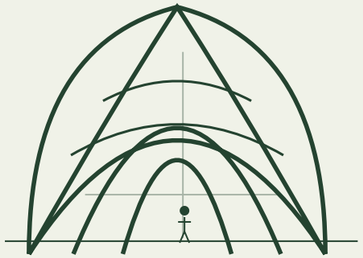
Garden room

02



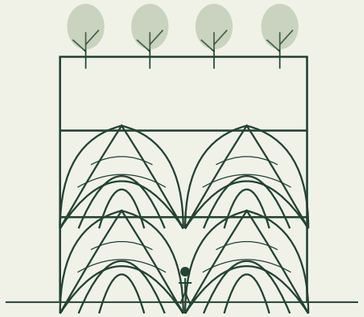
Cultural pavilion

03



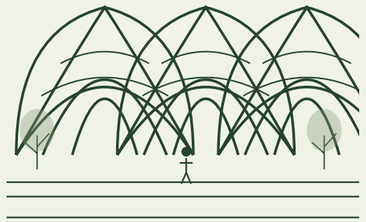
Sanctuary

04



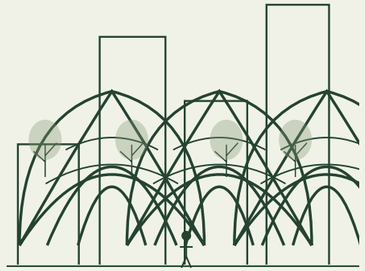
Living building

05



Planted street

06



Urban fabric

FIRST PRODUCT

LONG-TERM RESEARCH HORIZON

Long-term ambition. Speculative scale studies; engineering feasibility and lifecycle carbon performance remain to be demonstrated.

Buildings that *grow.*

We mine, pour, and cut; we assemble inert objects that begin decaying the moment they're complete; and we've quietly accepted that the largest thing our species makes — the built world — is also the most severed from the living one.

Make the first Bower possible.
Develop the means to make many.

bowerbuild.org/designx/5 ↗